

Boat and Stream

The chapter of boat and stream is based on the application of time and distance. Two terms which are frequently used in this chapter are downstream and upstream.

1. **Downstream** → When boat is moving along the direction of the stream.

2. **Upstream** → When boat is moving against the direction of the stream.

Basic formula: If 'x' be the speed of boat in still water and 'y' be the speed of stream.

(i) **Downstream speed (v) = (x + y) km/h**

(ii) **Upstream speed (u) = (x - y) km/h**

Where downstream Speed is 'v' and upstream speed is 'u'.

(iii) **Speed of boat in still water, (x) = $\frac{1}{2} (u + v)$**

(iv) **Speed of stream, (y) = $\frac{1}{2} (v - u)$**

Note: If the speed of water in river is zero, then the water is considered to be still.

Example: A man row with a speed of 8 km/h in still water. Find the downstream and upstream speed of boat, if the speed of stream is 4 km/h. ?

Sol. Downstream speed = (x + y) km/h = (8 + 4) = 12 km/h

Upstream speed = (x - y) km/h = (8 - 4) = 4 km/h

Concept 1: If the speed of boat in still water is x km/h and speed of stream is y km/h. If time taken to go and come back

from a point is T, the distance between both points = $\frac{T(x^2 - y^2)}{2x}$ km

Example: A man can row 14 km/h in still water. When the stream is running at 2 km/h, it takes him 7 hour to row to a place and to come back. How far is the place ?

Sol. Required distance = $\frac{7(14^2 - 2^2)}{2 \times 14}$ km = $\frac{7 \times 192}{2 \times 14}$ = 48 km

Concept 2: A person can row a certain distance downstream is t_1 h and returns upstream the same distance in t_2 h. When the stream flows at the rate of 's' km/h.

Then the speed of man = $\frac{s(t_1 + t_2)}{t_2 - t_1}$ km/h

Example: Sonu can row a certain distance is 8 h and can return the same distance is 12h. If the stream of flows at the rate of 8 km/h, then find the speed of Sonu is still water ?

Sol: Speed of man = $\frac{8(8 + 12)}{12 - 8}$ km/h = $\frac{8 \times 20}{4}$ = 40 km/h

Concept 3: If the speed of boat in still water is x km/h and river is flowing with a speed of y km/h, then average speed

in going to a certain place and coming back to starting point is given by $\frac{(x+y)(x-y)}{x}$ km/h.

Example: A man can row in still water with a speed of 6 km/h to go to a certain place and to come back. Find the average speed for the whole journey, if the river is flowing with a speed of 3 km/h ?

Sol. Average speed = $\frac{(6+3)(6-3)}{6}$ km/h = $\frac{9 \times 3}{6}$ km/h = 4.5 km/h

Concept 4: If a man can row d_1 km upstream and e_1 km downstream in T_1 hours. Also, he can row d_2 km upstream and e_2 km downstream in T_2 hours.

Then, the upstream speed of man = $\left(\frac{d_1 e_2 - d_2 e_1}{e_2 T_1 - e_1 T_2}\right)$ km/h $\Rightarrow$ Downstream speed of man = $\left(\frac{d_1 e_2 + d_2 e_1}{d_1 T_2 + d_2 T_1}\right)$ km/h

Example: A man can row 30 km upstream and 44 km downstream in 10 hrs. Also, he can row 40 km upstream and 55 km downstream in 13 hrs. Find the rate of current and speed of man in still water?

Sol. Upstream speed of man = $\frac{30 \times 55 - 40 \times 44}{55 \times 10 - 44 \times 13} = \frac{-110}{-22} = 5$ km/h

Downstream speed of man = $\frac{30 \times 55 + 40 \times 44}{30 \times 13 + 40 \times 10} = \frac{-110}{-10} = 11$ km/h

Speed of man = $\frac{5+11}{2} = 8$ km/h $\Rightarrow$ Speed of stream = $\frac{11-5}{2} = 3$ km/h

Types of Questions

1. A man can row upstream at 10 km/hr and downstream at 16 km/hr. Find the speed of man in still water?

Sol. Let the speed of the man in still water be x km/hr and speed of the stream be y km/hr.

According to the question,

$$x + y = 16 \quad \dots (i)$$

$$\text{and } x - y = 10 \quad \dots (ii)$$

Adding eq. (i) with eq. (ii), we have

$$2x = 26 \quad \therefore x = 13 \text{ km/hr}$$

$\therefore$ Speed of the man in still water = 13 km/hr.

2. A man can row upstream at 10 km/hr and downstream at 16 km/hr. Find the rate of the current?

Sol. Let the speed of the man in still water be x km/hr and the rate of the current be y km/hr

According to the question,

$$\text{Downstream speed} = x + y = 16 \text{ km/hr} \quad \dots (i)$$

$$\text{Upstream} = x - y = 10 \text{ km/hr} \quad \dots (ii)$$

Subtracting eq. (ii) from eq. (i), we have

$$2y = 6 \text{ km/hr} \quad \text{or } y = 3 \text{ km/hr}$$

$\therefore$ Speed of the current = 3 km/hr

3. A stream is running at 2 km/hr. A motorboat goes 10 km upstream and comes back to the starting point in 55 minutes. Find the speed of the motorboat in still water.

Sol. Let the speed of the motorboat in still water be x km/hr.

$$\frac{10}{x+2} + \frac{10}{x-2} = \frac{55}{60}$$

$$\text{or, } 240x = 11x^2 - 44$$

$$\text{or, } 11x^2 - 240x - 44 = 0$$

$$\therefore (x - 22)(11x + 2) = 0$$

So, $x = 22$ km/hr (neglect the -ve value)

$\therefore$ Speed of the motorboat in still water = 22 km/hr.

4. Ramesh can row a certain distance downstream in 6 hours and return the same distance in 9 hours. If the stream flows at the rate of 3 km per hour find the speed of Ramesh in still water.

Sol. Let the speed of Ramesh in still water be x km/hr.

Then his upstream speed = $(x - 3)$ km/hr

Downstream speed = $(x + 3)$ km/hr.

Now, we are given that up and down journey are equal, therefore,

$$(x + 3)6 = (x - 3)9 \quad \text{or, } 6x + 18 = 9x - 27$$

$$\text{or, } 3x = 45; \therefore x = 15 \text{ km/hr}$$

5. The speed of a boat in still water is 11 km/h and the speed of the stream is 3 km/h. The boat takes a total of 11 hours to cover a distance up and down of a fixed destination. What is the fixed distance?

Sol. Let the fixed distance be l km.

Speed of the boat downstream = $11 + 3 = 14$ km/h

Time taken to cover l km downstream = $\frac{l}{14}$ hours.

Now the speed of the boat upstream = $11 - 3 = 8$ km/h

Time taken to cover l km upstream = $\frac{l}{8}$ hours

From the question:

$$\frac{l}{14} + \frac{l}{8} = 11 \text{ or, } \frac{7l + 4l}{56} = 11 \quad \text{or, } l = 11 \times \frac{56}{11} = 56 \text{ km}$$

6. A steamer takes one hour more to go 48 km upstream than the time to go 60 km downstream. If the steamer takes equal time to cover 30 km downstream and 18 km upstream then find the ratio of the speed of the boat in still water and the speed of the stream.

Sol. Let the speed of the boat in still water = x km/h.

Speed of the stream is y km/h

So, The speed of steamer downstream = $x + y$ km/h

and the speed of the steamer upstream = $x - y$ km/h

$$\text{In first situation, } \frac{48}{x-y} - \frac{60}{x+y} = 1 \quad \dots (i)$$

and in second situation $\frac{30}{x+y} = \frac{18}{x-y}$

or, $\frac{60}{x+y} = \frac{36}{x-y}$... (ii)

from eq. (i) and (ii) $\frac{48}{x-y} - \frac{36}{x-y} = 1$

or, $\frac{12}{x-y} = 1$ or, $x - y = 12$... (iii)

Again from eq. (i), $\frac{48}{12} - \frac{60}{x+y} = 1$

or, $4 - 1 = \frac{60}{x+y}$ or, $x + y = 20$... (iv)

Solving eq. (iii) and (iv), $x = 16$ and $y = 4$

So required ratio = $\frac{x}{y} = \frac{16}{4} = 4 : 1$

Foundation

Questions

- A boat goes 14 km upstream in 42 minutes. The speed of stream is 4 km/h. The speed of boat in still water is
(a) 24 km/h (b) 25 km/h
(c) 30 km/h (d) 23 km/h
- The speed of a boat in still water is 7 km/h. If its speed in downstream is 10 km/h. Then the speed of stream is
(a) 7 km/h (b) 3 km/h
(c) 10 km/h (d) 6 km/h
- Aditya can row with stream at 10 km/h and against the stream at 6 km/h. His speed in still water is
(a) 16 km/h (b) 6 km/h
(c) 8 km/h (d) 10 km/h
- Speed of man is 12 km/h in still water. If the rate of current is 4 km/h, then the effective speed of the man upstream is
(a) 6 km/h (b) 7 km/h
(c) 5 km/h (d) 8 km/h
- A man moves downstream at a rate of 14 km/h and upstream at a rate of 6 km/h. Speed of boat in still water is
(a) 4 km/h (b) 10 km/h
(c) 16 km/h (d) 2 km/h
- A man moves 20 km down stream in 5 hours and 10 km up stream in same time. The speed of stream
(a) 4 km/h (b) 1 km/h
(c) 1.6 km/h (d) done
- A boat travels 18 km upstream in 6 hours. How long (In hrs) will it take to cover the same distance downstream. The speed of current is one fourth the speed of boat in still water.
(a) 3 (b) 4
(c) 3.6 (d) 3.4
- A person can row with the stream at 8 km/h and against the stream at 4 km/h. The speed of the current is (In km/hr)
(a) 1 (b) 2
(c) 1.5 (d) 4
- When the speed of a boat in still water is 4 km/h and the rate of stream is 2 km/h, find upstream speed of the boat?
(a) 6 km/h (b) 5 km/h
(c) 2 km/h (d) 7 km/h
- If the speed of a swimmer in still water is 9 km/h. Find the downstream speed of the swimmer, when the river is flowing with the speed of 6 km/h?
(a) 15 km/h (b) 18 km/h
(c) 3 km/h (d) 12 km/h
- A swimmer's speed in downstream is 11 km/h and speed of the stream is 1.5 km/h. Find the upstream speed of swimmer?
(a) 8 km/h (b) 9.5 km/h
(c) 9 km/h (d) 6.25 km/h
- A boatman rows 1 km in 5 min along the stream and 6 km in 1 h against the stream. The speed of the stream is
(a) 3 km/h (b) 6 km/h
(c) 10 km/h (d) 12 km/h
- Keshav can row 60 km downstream and 36 km upstream, taking 10 h each time. What is the velocity of the current?
(a) 3 km/h (b) 2.2 km/h
(c) 4 km/h (d) 1.2 km/h
- What time will be taken by a boat to cover a distance of 128 km along the stream, if speed of a boat in still water is 24 km/h and speed of stream is 8 km/h?
(a) 8 h (b) 4 h
(c) 7 h (d) 6 h
- A man can row against the current three-fourth of a kilometre in 15 min and returns same distance in 10 min, then ratio of his speed to that of current is -
(a) 3 : 5 (b) 5 : 3
(c) 1 : 5 (d) 5 : 1
- A boat goes 48 km downstream in 20 h. It takes 4 h more to cover the same distance against the stream. What is the speed of the boat in still water?
(a) 2.2 km/h (b) 2 km/h
(c) 4 km/h (d) 4.2 km/h

17. A man can row 15 km downstream in 3 hours and 5 km upstream in $2\frac{1}{2}$ hours. His speed in still water in (km/hr) is
 (a) 4 km/hr (b) 4.5 km/hr
 (c) 3.5 km/hr (d) 5 km/hr
18. A man can row 44 km downstream in 4 hours. If the man's rowing rate in still water is 8 km/hr, then find in what time will he cover 25 km upstream?
 (a) 5 hours (b) 6 hours
 (c) 4.5 hours (d) 4 hours
19. A man can row 20 km/h in still water and speed of the stream is 5 m/sec. Ratio of speed of man to current speed is
 (a) 9 : 10 (b) 4 : 1
 (c) 1 : 4 (d) 10 : 9
20. A man can swim in still water at 9.6 km/hr, but takes twice as long to swim upstream then downstream. The speed of stream is
 (a) 3.2 km/hr (b) 2.3 km/hr
 (c) 5.6 km/hr (d) 6.5 km/hr

Moderate

1. Speed of motorboat in still water is 45 km/h. If the motorboat travels 80 km along the stream in 1 h 20 min, then the time taken by it to cover the same distance against the stream will be
 (a) 4 h 20 min (b) 3 h 40 min
 (c) 2 h 40 min (d) 2 h 55 min
2. Speed of boat in still water is 5 km/h. The river is flowing with a speed of 2 km/h and time taken to cover a certain distance upstream is 2 h more than time taken to cover the same distance downstream. Find the distance?
 (a) 10.5 km (b) 11 km
 (c) 10.9 km (d) 15 km
3. A man can row at 10 km/h in still water. If he takes total 5 h to go to a place 24 km away and return, then the speed of the water current is –
 (a) 2 km/h (b) 3 km/h
 (c) $\frac{1}{2}$ km/h (d) 1 km/h
4. A steamer goes downstream from one port to another in 4 h. It covers the same distance upstream in 5 h. If the speed of the stream is 2 km/h, then find the distance between the two ports.
 (a) 50 km (b) 60 km
 (c) 70 km (d) 80 km
5. A boatman takes twice as long to row a distance against the stream as to row the same distance with the stream. Find the ratio of speeds of the boat in still water and the stream?
 (a) 2 : 1 (b) 3 : 1
 (c) 1 : 2 (d) 1 : 3
6. A river is flowing with a steady speed of 4 km/h. One rows his boat downstream in the river and then returns by rowing upstream in the same river. When he returns to the starting point, the total distance covered by him is 42 km. If the return journey takes 2 h more than his onward journey, then the speed of his rowing in still water must be:
 (a) 12 km/h (b) 10 km/h
 (c) 9 km/h (d) 8 km/h
7. The ratio of speeds of a motorboat to that of the current of water is 36 : 5. The motorboat goes along with the current in 5 h 10 min. Find the time to come back of motorboat.
 (a) 5 h 50 min (b) 6 h
 (c) 6 h 50 min (d) 12 h 10 min
8. In a river, the ratio of the speed of stream and speed of a boat in still water is 2 : 5. Again, ratio of the speed of stream and speed of another boat in still water is 3 : 4. What is the ratio of the speeds of the first boat to the second boat in still water?
 (a) 10 : 7 (b) 15 : 8
 (c) 4 : 3 (d) 5 : 4
9. A man can row at the rate of 3.5 km/hr in still water. If the time taken to row a certain distance upstream is $2\frac{1}{2}$ times as much as to row the same distance downstream, find the speed of the current.
 (a) 2.5 km/hr (b) 1.5 km/hr
 (c) 3 km/hr (d) 1.25 km/hr
10. A man can row at the rate of 4 km/hr in still water. If the time taken to row a certain distance upstream is 3 times as much as to row the same distance downstream, find the speed of the current?
 (a) 3 km/hr (b) 1.5 km/hr
 (c) 1 km/hr (d) 2 km/hr
11. A man swimming in a stream which flows 1.5 km/hr finds that in a given time he can swim twice as far with the stream as he can against it, at what rate does he swim?
 (a) 4 km/hr (b) 4.5 km/hr
 (c) 5 km/hr (d) 3.5 km/hr
12. A man can row upstream 36 km in 6 hours. If the speed of a man in still water is 8 km/hr, find how much he can go downstream in 10 hours?

- (a) 150 km (b) 80 km
(c) 90 km (d) 100 km
13. The speed of a boat in still water is 4 km/hr and the speed of current is 2 km/hr. If the time taken to reach a certain distance upstream is 9 hours, find the time it will take to go to same distance downstream ?
(a) 2 hrs (b) 2.5 hrs
(c) 3.5 hrs (d) 3 hrs
14. A person can swim in still water at 4 km/hr. If the speed of water is 2 km/hr, how many hours will the man take to swim back against the current for 6 km ?
(a) 3 (b) 4
(c) $4\frac{1}{2}$ (d) 5
15. A man can row at a speed of 4.5 km/hr in still water to a certain upstream point and back to the starting point in a river which flows at 1.5 km/hr. Find his average speed for total journey?
(a) 4 km/hr (b) 6 km/hr
(c) 4.5 km/hr (d) 5 km/hr
16. A man can row at a speed of 15 km/hr in still water to a certain upstream point and back to the starting point in a river which flows at 3 km/hr. Find his average speed for total journey ?
(a) 9 km/hr (b) 6 km/hr
(c) 3 km/hr (d) 14.4 km/hr
17. A man can row 7 km/hr in still water. If the river is running at 3 km/hr, it takes 6 hours more in upstream than to go downstream for the same distance. How far is the place?
(a) 48 km (b) 36 km
(c) 42 km (d) 40 km
18. A man can row 9 km/hr in still water. If the river is running at 3 km/hr, it takes 3 hours more in upstream than to go downstream for the same distance. How far is the place?
(a) 30 km (b) 36 km
(c) 24 km (d) 42 km
19. A boat takes 3 hours to travel from place M to N downstream and back from N to M upstream. If the speed of boat in still water is 4 km/hr, what is the distance between the two places?
(a) 8 km (b) 12 km
(c) 6 km (d) Data inadequate
20. P, Q and R are the three towns on a river which flows uniformly. Q is equidistant from P and R. I row from P to Q and back in 10 hours and I can row downstream from P to R in 4 hours. Compare the speed of my boat in still water with that of the river.
(a) 5 : 3 (b) 4 : 3
(c) 6 : 5 (d) 7 : 3

Difficult

1. A motorboat travelling at some speed, can cover 25 km upstream and 39 km downstream in 8 h. At the same speed, it can travel 35 km upstream and 52 km downstream in 11 h. The speed of the stream is:
(a) 2 km/h (b) 3 km/h
(c) 4 km/h (d) 5 km/h
2. A man can row a boat at a speed of 8 kmph in still water. He was rowing the boat downstream from one point to another. After travelling half of the distance the motor of the boat failed and stopped working. He travelled on the boat along the stream and reached his destination taking six hours more than the usual time. The speed of the stream is 2 kmph. What is the distance between the two points?
(a) 20 (b) 30
(c) 24 (d) 28
3. A, B and C are situated at the bank of river which is flowing at a constant rate. B is at an equal distance with A and C. A swimmer Avinash takes 10 h to swim from A to B and B to A. Also, he takes 8 h to swim from A to C in downstream. What is the ratio of speed of Avinash in still water and speed of stream?
(a) 5 : 4 (b) 3 : 5
(c) 5 : 1 (d) 1 : 2
4. The speed of a boat downstream is $\frac{16}{9}$ times the speed of the boat upstream. The speed of the current is what percent of the speed of the boat in still water?
(a) 28% (b) 24%
(c) 56.25% (d) 40%
5. A river is flowing at a speed of 5 km/h in a particular direction. A man, who can swim at a speed of 20 km/h in still water, starts swimming along the direction of flow of the river from point A and reaches another point B which is at a distance of 30 km from the starting point A. On reaching point B, the man turns back and starts swimming against the direction of flow of the river and stops after reaching point A. The total time taken by the man to complete his journey is
(a) 2 h 30 min (b) 3 h 12 min
(c) 3 h 30 min (d) 3 h 45 min
6. A boat covered a certain distance travelling downstream in 45 minutes, while it came back to the starting point in one hour 15 minutes. The speed of the stream is 2 kmph. What is the speed of the boat in still water?
(a) 8 kmph (b) 9 kmph
(c) 10 kmph (d) 11 kmph

7. In a stream running at 2 km/hr, a water scooter goes 10 km upstream and back again to the starting point in 55 minutes. Find the speed of the water scooter in still water.
 (a) 22 km/hr (b) 26 km/hr
 (c) 28 km/hr (d) 30 km/hr
8. A man can row 40 km upstream and 55 km downstream in 13 hours. Also, he can row 30 km upstream and 44 km downstream in 10 hours. Find the speed of the man in still water and the speed of the current.
 (a) 8 kmph & 3 kmph (b) 6 kmph & 2 kmph
 (c) 10 kmph & 4 kmph (d) Can not be determined
9. A person takes total 5 hours to go the destination downstream and also comes back upstream by boat. If the speed of the boat in still water is 10 km/h and the speed of the stream is 4 km/h, then what is the distance of destination from the starting point?
 (a) 16 km (b) 18 km
 (c) 21 km (d) 25 km
10. A boat covers 24 km upstream and 36 km downstream in 6 hours while it covers 36 km upstream and 24 km downstream in $6\frac{1}{2}$ hours. The velocity of the current is:
 (a) 2 km/hr (b) 3 km/hr
 (c) 4 km/hr (d) 5 km/hr
11. The speed of boat in still water is 10 km/hr. If it can travel 26 km downstream and 14 km upstream in the same time, the speed of the stream is?
 (a) 2 km/hr (b) 2.5 km/hr
 (c) 3 km/hr (d) 4 km/hr
12. In a stream running at 2 kmph, a motorboat goes 6 km upstream and back again to the starting point in 33 minutes. Find the speed of the motorboat in still water?
 (a) 22 (b) 23
 (c) 20 (d) 25
13. A man row $9\frac{1}{3}$ kmph in still water and finds that it takes him thrice as much time to row up than as to row down the same the same distance in the river. The speed of the current is:
 (a) $3\frac{1}{3}$ km/hr (b) $3\frac{1}{9}$ km/hr
 (c) $4\frac{2}{3}$ km/hr (d) $4\frac{1}{2}$ km/hr
14. A man can row 18 kmph in still water, it takes him thrice as long to row up as to row down the river. Find the rate of stream.
 (a) 8 (b) 9
 (c) 10 (d) 11
15. Rohit can row a certain distance downstream in 8 hours and return the same distance in 12 hours. If the stream flows at the rate of 5 km per hour. Find the speed of Rohit in still water?
 (a) 20 km/hr (b) 30 km/hr
 (c) 15 km/hr (d) 25 km/hr
16. A man takes 3 hours 45 minutes to row a boat 15 km downstream of a river and 2 hours 30 minutes to cover a distance of 5 km upstream. find the speed of the river current in km/hr.
 (a) 1 (b) 2
 (c) 3 (d) 4
17. A boat goes 6 km upstream and back again to the starting point in 2 hours. If the velocity of the stream is 4 km/hr, what is the speed of the boat in still water?
 (a) 6.5 km/hr (b) 7.5 km/hr
 (c) 8 km/hr (d) 8.5 km/hr
18. Speed of a boat in standing water is 9 kmph and the speed of the stream is 1.5 kmph. A man rows to a place at a distance of 105 km and comes back to the starting point. The total time taken by him is:
 (a) 16 (b) 18
 (c) 20 (d) 24
19. A boat covers 12 km upstreram and 18 km downstream in 3 hours while it covers 36 km upstream and 24 km downstream in $6\frac{1}{2}$ hours, what is the velocity of the stream?
 (a) 1.5 km/hr (b) 1 km/hr
 (c) 2 km/hr (d) 2.5 km/hr
20. A man rows to a place 48 km distance and come back in 14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. The rate of the stream is:
 (a) 1 (b) 1.5
 (c) 2 (d) 2.5

Previous Year (Memory Based)

1. A boat goes 11 km along the stream in 1 hr and 5 km against the stream in 1 hr. The speed of the boat in still water (in km/hr) is:
 (a) 3 (b) 5
 (c) 8 (d) 9
2. A boat running downstream covers distance of 16 km in 2 hours while for covering the same distance upstream, it takes 4 hours. What is the speed of the boat in still water?
 (a) 4 km/hr (b) 6 km/hr
 (c) 8 km/hr (d) Data inadequate
3. A boatman goes 2 km against the current of the stream in 1 hour and goes 1 km along the current in 10 minutes. How long will it take to go 5 km in stationary water?
 (a) 40 minutes (b) 1 hour
 (c) 1 hr 15 min (d) 1 hr 30 min
4. A man takes twice as long to row a distance against the stream as to row the same distance in favour of the stream. The ratio of the speed of the boat (in still water) and the stream is:
 (a) 2 : 1 (b) 3 : 1
 (c) 3 : 2 (d) 4 : 3
5. A boat running upstream takes 8 hours 48 minutes to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and speed of the water current respectively?
 (a) 2 : 1 (b) 3 : 2
 (c) 8 : 3 (d) Cannot be determined
6. If a boat goes 7 km upstream in 42 minutes and the speed of the stream is 3 kmph, then the speed of the boat in still water is:
 (a) 4.2 km/hr (b) 9 km/hr
 (c) 13 km/hr (d) 21 km/hr
7. A man can row at 5 kmph in still water. If the velocity of current is 1 kmph and it takes him 1 hour to row to a place and come back, how far is the place?
 (a) 2.4 km (b) 2.5 km
 (c) 3 km (d) 3.6 km
8. A motorboat, whose speed is 15 km/hr in still water goes 30 km downstream and comes back in a total of 4 hours 30 minutes. The speed of the stream (in km/hr) is:
 (a) 4 (b) 5
 (c) 6 (d) 10
9. A boat takes 90 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 mph, the speed of the stream is:
 (a) 2 mph (b) 2.5 km/hr
 (c) 3 mph (d) 4 mph
10. At his usual rowing rate, Rahul can travel 12 miles downstream in a certain river in 6 hours less than it takes him to travel the same distance upstream. But if he could double his usual rowing rate for his 24 mile round trip, the downstream 12 miles would then take only one hour less than the upstream 12 miles. What is the speed of the current in miles per hour?
 (a) $1\frac{1}{3}$ (b) $1\frac{2}{3}$
 (c) $2\frac{1}{3}$ (d) $2\frac{2}{3}$
11. A motorboat went downstream for 28 km and immediately returned. It took the boat twice as long to make the return trip. If the speed of the river flow were twice as high, the trip downstream and back would take 672 minutes. Find the speed of the boat in still water and the speed of the river flow:
 (a) 9 km/h, 3 km/h (b) 9 km/h, 6 km/h
 (c) 8 km/h, 2 km/h (d) 12 km/h, 3 km/h
12. Two friends A and B, on their last day in college, decided to meet after 20 years on a river. A had to sail 42 km to the meeting place and B had to sail $35\frac{5}{7}$ per cent less. To arrive at the meeting place at the same as his friend B, A started at the same time as B and sailed with the speed exceeding by 5 km/h the speed of B. Find the speed of A:
 (a) 10 kmph (b) 14 kmph
 (c) 9 kmph (d) 21 kmph
13. On the banks of the river Yamuna there are two bathing points in Varanasi and Patna. A diya left in the river at Varanasi reaches Patna in 24 hours. However, a motorboat covers the whole way to and fro in exactly 10 hours. If the speed of the motorboat in still water increased by 40% then it takes the motor boat 7 hours to cover the same way (from Varanasi to Patna and back again). Find the time necessary for the motorboat to sail from Varanasi to Patna when its speed in still water is not increased:
 (a) 3 hours (b) 4 hours
 (c) 4.8 hours (d) None of these
14. A motorboat moves from point A to point B and back again, both points being located on the riverbank. If the speed of the boat in still water is doubled, then the trip from A to B and back again would take 20% of the time that the motorboat usually spends in the journey. How many times is actual speed of the motor boat higher than the speed of the river flow?

$$(a) \sqrt{\frac{3}{2}} \quad (b) \sqrt{\frac{2}{3}}$$

$$(c) \frac{2}{3} \quad (d) \frac{3}{2}$$

15. The normal speed of a boat in still water is 4 times the speed of the river. The boat goes upstream and takes 2 hrs to reach B from A. By what percent should the boat increase/decrease its speed if it is required to reach A from B in exactly 1 hr 30 minutes?

- (a) increase by 20% (b) increase by 25%
(c) decrease by 20% (d) decrease by 25%

16. Two boats start at the same instant to cross a river W metre wide. The faster boat reaches the other bank and returns immediately. What are the distance travelled by them when they meet, where the speeds of these boats are b_1 & b_2 ?

$$(a) \frac{2W}{(b_1 + b_2)}, \frac{2W}{(b_1 - b_2)}$$

$$(b) \frac{2W}{(b_1 + b_2)} b_1 \text{ and } \frac{2W}{(b_1 + b_2)} b_2$$

$$(c) \frac{2W}{(b_1 + b_2)} b_2$$

- (d) data insufficient

17. A boat goes 60 km and back to starting point in 10 hours. The time taken by the boat to row 3 km downstream is equal to the time taken by the boat to row 2 km upstream. Find the speed of boat in still water and rate of current:

- (a) 12.5 km/hr, 2.5 km/hr
(b) 15 km/hr, 10 km/hr
(c) 15 km/hr, 12 km/hr
(d) None of these

18. A man can row against the current $\frac{3}{4}$ of a kilometre in 15 min and returns the same distance in 10 min. The ratio of this speed to that of the current is:

- (a) 3 : 5 (b) 5 : 3
(c) 1 : 5 (d) 5 : 1

19. The ratio of speed of a motor-boat to that of the current of water is 36 : 5. The boat goes along with the current for 10 minutes. It will come back in:

- (a) 12 min (b) 14.5 min
(c) 13.22 min (d) 11.6 min

20. A boat goes 20 km downstream in 1h and the same distance upstream in 2h. The speed of the boat in still water is:

- (a) 15 km/h (b) 10 km/h
(c) 5 km/h (d) 7.5 km/h

Foundation

Solutions

1. (a); Let speed of boat in still water = x

$$\text{Then, } \frac{14}{x-4} = \frac{42}{60} \Rightarrow x = 24 \text{ km/h}$$

2. (b); Let the speed of the stream is y km/h.

$$\text{Then, } y + 7 = 10 \Rightarrow y = 3 \text{ km/h}$$

3. (c); Let speed of boat = x km/h

$$\text{Speed of stream} = y \text{ km/h}$$

$$\text{Then, } x + y = 10 \Rightarrow x - y = 6$$

$$\Rightarrow 2x = 16 \Rightarrow x = 8 \text{ km/h}$$

4. (d); Speed of the man upstream (v) = $12 - 4 = 8$ km/h

5. (b); Let speed of boat in still water = x km/hr

$$\text{Speed of stream} = y \text{ km/hr}$$

$$x + y = 14, \quad x - y = 6,$$

$$2x = 20, \quad x = 10 \text{ km/h}$$

6. (b); Let speed of boat in still water = x

$$\text{Speed of stream} = y$$

$$\frac{20}{x+y} = 5 \Rightarrow x + y = 4 \quad \dots (i)$$

$$\frac{10}{x-y} = 5 \Rightarrow x - y = 2 \quad \dots (ii)$$

$$\text{From (i) and (ii), } 2x = 6 \Rightarrow x = 3$$

$$y = 1 \text{ km/h}$$

7. (c); Let speed of boat in still water = x

$$\text{From question, Given } y = \frac{1}{4}x$$

$$\text{Then } \frac{18}{x - \frac{x}{4}} = 6 \Rightarrow 3 = \frac{3x}{4} \Rightarrow x = 4$$

Then the time it will take to cover the same

$$\text{distance downstream} = \frac{18}{4+1} = \frac{18}{5} = 3.6 \text{ hr.}$$

8. (b); Let the speed of boat = x

$$\text{Speed of stream} = y$$

$$x + y = 8 \text{ km/h} \Rightarrow x - y = 4 \text{ km}$$

$$2x = 12 \Rightarrow x = 6 \Rightarrow y = 2 \text{ km/h}$$

9. (c); Given $x = 4$ km/h $y = 2$ km/h

$$\text{Upstream rate} = 4 - 2 = 2 \text{ km/h}$$

10. (a); Given $x = 9, y = 6$

Then downstream speed of the swimmer
 $= 9 + 6 = 15 \text{ km/h}$

11. (a); Given, $x + y = 11 \Rightarrow y = 1.5$ then $x = 9.5$
 $\therefore$ upstream speed of swimmer $= 9.5 - 1.5$
 $= 8 \text{ km/hr}$

12. (a); Let speed of boat = x , Speed of steam = y

$$\frac{1}{x+y} = \frac{5}{60} \Rightarrow x+y = 12 \quad \dots (i)$$

$$\frac{6}{x-y} = 1 \Rightarrow x-y = 6 \quad \dots (ii)$$

From (i) and (ii)

$$2x = 18 \Rightarrow x = 9$$

Speed of the stream $= (12 - 9) = 3 \text{ km/h}$

13. (d); Let speed of boat = x , speed of current = y

$$\frac{60}{x+y} = 10 \Rightarrow x+y = 6 \quad \dots (i)$$

$$\frac{36}{x-y} = 10 \Rightarrow x-y = 3.6 \quad \dots (ii)$$

From (i) and (ii)

$$2x = 9.6 \Rightarrow x = 4.8 \Rightarrow y = 1.2 \text{ km/h}$$

14. (b); Time taken $= \frac{128}{24+8} = \frac{128}{32} = 4 \text{ h}$

15. (d); Let speed of boat = x , speed of stream = y

$$\frac{3/4}{x-y} = \frac{15}{60} \Rightarrow x-y = 3 \quad \dots (i)$$

$$\frac{3/4}{x+y} = \frac{10}{60} \Rightarrow \frac{3}{4(x+y)} = \frac{1}{6}$$

$$\Rightarrow x+y = \frac{9}{2} \quad \dots (ii)$$

From (i) and (ii)

$$2x = 3 + \frac{9}{2} \Rightarrow 2x = \frac{15}{2} \Rightarrow x = \frac{15}{4}$$

$$\text{and } y = \frac{15}{4} - 3 \Rightarrow y = \frac{15-12}{4} = y = \frac{3}{4}$$

$$\therefore \frac{x}{y} = \frac{15/4}{3/4} = \frac{15}{3} = \frac{5}{1} \Rightarrow x:y = 5:1$$

16. (a); Let speed of boat = x , speed of stream = y

$$\frac{48}{x+y} = 20 \Rightarrow x+y = \frac{12}{5} \quad \dots (i)$$

$$\frac{48}{x-y} = 24 \Rightarrow x-y = 2 \quad \dots (ii)$$

From (i) and (ii)

$$2x = \frac{12}{5} + 2 \Rightarrow x = \frac{11}{5}$$

$$x = 2.2 \text{ km/h}$$

17. (c); Let speed of boat = x , speed of stream = y

$$\frac{15}{x+y} = 3 \Rightarrow x+y = 5 \quad \dots (i)$$

$$\frac{5}{x-y} = \frac{5}{2} \Rightarrow x-y = 2 \quad \dots (ii)$$

From (i) and (ii)

$$x = 3.5 \text{ km/h}$$

18. (a); Let speed of the current = $y \text{ km/h}$

$$\text{Given } \frac{44}{8+y} = 4 \Rightarrow y = 3 \text{ km/h}$$

$$\text{Time taken in upstream} = \frac{25}{(8-3)} = 5 \text{ h}$$

19. (d); Speed in still water = 20 km/h

$$\text{Speed of the stream} = 5 \text{ m/sec} = 5 \times \frac{18}{5} \text{ km/hr}$$

$$= 18 \text{ km/h} \Rightarrow \text{Ratio} = 20 : 18 = 10 : 9$$

20. (a); Given speed in still water (x) = 9.6 km/h

Let speed of stream = $y \text{ km/h}$, distance = d

$$\text{According to question } \frac{d}{(9.6-y)} = \frac{2d}{(9.6+y)}$$

$$2 \times 9.6 - 2y = 9.6 + y \Rightarrow 3y = 9.6$$

$$y = 3.2 \text{ km/h}$$

Moderate

1. (c); Speed of motor boat in still water = 45 km/h

Time taken to travel along the stream

$$= 1 \text{ hr } 20 \text{ min} = 1 \frac{20}{60} = 1 \frac{1}{3} = \frac{4}{3} \text{ hr.}$$

Let speed of current = $y \text{ km/h}$

Then According to question,

$$\frac{80}{45+y} = \frac{4}{3} \Rightarrow 60 = 45 + y \Rightarrow y = 15 \text{ km/h}$$

$$\text{Then required time} = \frac{80}{45-15} = \frac{80}{30}$$

$$= \frac{80}{30} \times 60 \text{ min} = 160 \text{ min} = 2 \text{ hr } 40 \text{ min.}$$

2. (a); Speed of boat in still water = 5 km/h
speed of current = 2 km/h
Let distance = d

$$\frac{d}{5-2} = 2 + \frac{d}{5+2} \Rightarrow \frac{d}{3} = 2 + \frac{d}{7}$$

$$\frac{d}{3} - \frac{d}{7} = 2 \Rightarrow d = 10.5 \text{ km.}$$

3. (a); Speed of boat in still water = 10 km/h
Let speed of current = y km/h

$$\frac{24}{10+y} + \frac{24}{10-y} = 5 \Rightarrow 24 \left(\frac{10+y+10-y}{(10+y)(10-y)} \right) = 5$$

$$\frac{24 \times 20}{100-y^2} = 5 \Rightarrow 100 - y^2 = 96 \Rightarrow y^2 = 4$$

$$y = 2 \text{ (Neglect -ve sign)}$$

4. (d); Speed of the stream = 2 km/h
Let speed of the boat in still water = x km/h
And distance b/w two boats = d km

$$\text{Then } \frac{d}{x+2} = 4 \Rightarrow d - 4x = 8 \quad \dots (i)$$

$$\frac{d}{x-2} = 5 \Rightarrow d - 5x = -10 \quad \dots (ii)$$

From eq. (i) and (ii)

$$d - 4x - d + 5x = 8 + 10 \Rightarrow x = 18$$

From eq. (i)

$$d - 18 \times 4 = 8 \Rightarrow d = 80 \text{ km}$$

5. (b); Let speed of boat in still water = x
speed of current = y $\Rightarrow$ distance = d

$$\text{Then } \frac{d}{x-y} = \frac{2d}{x+y} \Rightarrow 2x - 2y = x + y$$

$$x = 3y \Rightarrow \frac{x}{y} = \frac{3}{1}$$

Required ratio = 3 : 1

6. (b); Given speed of current = 4 km/h
Let speed of rowing in still water = x km/h

$$\text{distance (d)} = \frac{42}{2} = 21 \text{ km}$$

$$\text{Then } \frac{21}{x-4} = 2 + \frac{21}{x+4}$$

$$21 \left(\frac{1}{x-4} - \frac{1}{x+4} \right) = 2 \Rightarrow 21 \left(\frac{8}{x^2-16} \right) = 2$$

$$21 \times 8 = 2x^2 - 32 \Rightarrow 200 = 2x^2$$

$$x^2 = 100 \Rightarrow x = 10 \text{ km/h}$$

7. (c); Let the speed of the motorboat = 36 x
Speed of current = 5x
let distance = d

$$\frac{d}{41x} = 5 \frac{10}{60} = \frac{31}{6} \Rightarrow \frac{d}{31x} = \frac{41}{6}$$

$$\text{Means, time to come back of motor boat} = \frac{41}{6} \text{ hr}$$

$$= \frac{41}{6} \times 60 \text{ min} = 410 \text{ min} = 6 \text{ h } 50 \text{ min}$$

8. (b); Let speed of first boat = 5a, stream = 2a
Speed of stream = 3b, boat = 4b

$$\text{Stream should be same} \Rightarrow 2a = 3b \Rightarrow a = \frac{3}{2}b$$

$$\text{Required Ratio} = \frac{5a}{4b} = \frac{5 \times 3}{2 \times 4} = \frac{15}{8}$$

9. (b); x = 3.5 km/h Let speed of current = y

$$\frac{d}{x-y} = \frac{5}{2} \times \frac{d}{x+y} \Rightarrow 5x - 5y = 2x + 2y$$

$$3x = 7y \Rightarrow 3 \times 3.5 = 7y \Rightarrow y = 1.5 \text{ km/h}$$

10. (d); Given x = 4 km/h. Let speed of current = y

$$\frac{d}{x-y} = \frac{3d}{x+y} \Rightarrow 3x - 3y = x + y$$

$$2x = 4y \Rightarrow 8 = 4y \Rightarrow y = 2 \text{ km/h}$$

11. (b); Given y = 1.5 km/h

Let speed of swimmer in still water = x

$$\frac{d}{x-y} = \frac{2d}{x+y} \Rightarrow x + y = 2x - 2y$$

$$x = 3y \Rightarrow x = 3 \times 1.5 \Rightarrow x = 4.5 \text{ km/hr}$$

12. (d); x = 8 km/h, Let speed of current = y

$$\frac{36}{x-y} = 6 \Rightarrow \frac{36}{8-y} = 6 \Rightarrow 36 = 48 - 6y$$

$$6y = 12 \Rightarrow y = 2 \Rightarrow \frac{d}{8+2} = 10 \Rightarrow d = 100 \text{ km.}$$

13. (d); x = 4 km/h, y = 2 km/h

$$\frac{d}{4-2} = 9 \Rightarrow d = 18 \text{ km} \Rightarrow \frac{18}{4+2} = t \Rightarrow t = 3 \text{ hr}$$

14. (a); x = 4 km/h y = 2 km/hr

$$\frac{6}{4-2} = t \Rightarrow t = 3 \text{ hr}$$

15. (a); x = 4.5 km/hr y = 1.5 km/hr

upstream speed (u) = 4.5 - 1.5 = 3 km/h

downstream = (v) = 4.5 + 1.5 = 6 km/h

$$\text{Average speed} = \frac{2uv}{u+v} = \frac{2 \times 3 \times 6}{9} = 4 \text{ km/hr}$$

16. (d); $x = 15 \text{ km/h}$, $y = 3 \text{ km/hr}$
 $u = 12 \text{ km/h}$, $v = 18 \text{ km/h}$

$$\text{Average speed} = \frac{2uv}{u+v} = \frac{2 \times 18 \times 12}{30}$$

$$= 14.4 \text{ km/hr}$$

17. (d); $x = 7 \text{ km/hr}$, $y = 3 \text{ km/hr}$

$$\frac{d}{7-3} = 6 + \frac{d}{10} \Rightarrow \frac{d}{4} = 6 + \frac{d}{10} \Rightarrow \frac{3d}{20} = 6$$

$$d = 40 \text{ km}$$

18. (b); $x = 9$, $y = 3$

$$\frac{d}{9-3} = 3 + \frac{d}{9+3} \Rightarrow \frac{d}{6} = 3 + \frac{d}{12} \Rightarrow d = 36 \text{ km}$$

19. (d); $x = 4 \text{ km/h}$

$$\frac{d}{4+y} + \frac{d}{4-y} = 3$$

We can't solve this question.

20. (a); $\frac{d}{x+y} + \frac{d}{x-y} = 10$... (i)

$$\frac{2d}{x+y} = 4 \Rightarrow d = 2(x+y)$$
 ... (ii)

From equation (i)

$$d \left(\frac{x-y+x+y}{(x+y)(x-y)} \right) = 10 \Rightarrow 2xd = 10(x+y)(x-y)$$

$$xd = 5(x+y)(x-y)$$

From equation (ii) put $(x+y) = \frac{d}{2}$

$$xd = 5 \times \frac{d}{2} (x-y) \Rightarrow 2x = 5x - 5y \Rightarrow 3x = 5y$$

$$3x = 5y \Rightarrow \frac{x}{y} = \frac{5}{3}$$

Difficult

1. (c); $\frac{25}{x-y} + \frac{39}{x+y} = 8$... (i)

$$\frac{35}{x-y} + \frac{52}{x+y} = 11$$
 ... (ii)

Let $x-y = A$, $x+y = B$

$$\frac{25}{A} + \frac{39}{B} = 8$$
 ... (iii)

$$\frac{35}{A} + \frac{52}{B} = 11$$
 ... (iv)

Eq. (iii) $\times 7$ - eq. (iv) $\times 5$

$$\left(\frac{175}{A} + \frac{273}{B} = 56 \right) - \left(\frac{175}{A} + \frac{260}{B} = 55 \right)$$

$$B = 13$$

Put it in eq. (iii)

$$\frac{25}{A} + 3 = 8 \Rightarrow A = 5 \Rightarrow x-y = 5$$
 ... (a)

$$x+y = 13$$
 ... (b)

From equation (a) and (b)

$$x = 9, y = 4$$

Speed of stream = 4 km/hr.

2. (b); Let d be the distance between two points

$$\frac{(d/2)}{10} + \frac{(d/2)}{2} = \frac{d}{10} + 6$$

$$\frac{d}{20} + \frac{d}{4} - \frac{d}{10} = 6 \Rightarrow d = 6 \left(\frac{20}{4} \right) = 30 \text{ km}$$

3. (c); $\frac{d}{x+y} + \frac{d}{x-y} = 10$... (i)

$$\frac{2d}{x+y} = 8 \Rightarrow 4(x+y) = d$$
 ... (ii)

From (i) $d \times \frac{x-y+x+y}{(x+y)(x-y)} = 10$

$$d \times 2x = 10(x+y)(x-y)$$

$$xd = 5(x+y)(x-y)$$

From (ii), $xd = 5 \times \frac{d}{4} \times (x-y) \Rightarrow x = \frac{5}{4}x - \frac{5}{4}y$

$$x = 5y \Rightarrow \frac{x}{y} = \frac{5}{1}$$

4. (a); $(x+y) = \frac{16}{9}(x-y) \Rightarrow 9x+9y = 16x-16y$

$$7x = 25y \Rightarrow y = \frac{7x}{25}$$

$$\text{Required percentage} = \frac{7x}{25x} \times 100 = 28\%$$

5. (b); Given, $x = 20 \text{ km/h}$, $y = 5 \text{ km/h}$

$$\text{Total time taken} = \frac{30}{25} + \frac{30}{15} = 2 + \frac{6}{5}$$

$$= \frac{16}{5} \text{ hr} = \frac{16}{5} \times 60 \text{ min} = 192 \text{ min} = 3 \text{ h } 12 \text{ min}$$

$$6. (a); \frac{d}{x+2} = \frac{3}{4} \Rightarrow 4d = 3x + 6 \quad \dots (i)$$

$$\frac{d}{x-2} = \frac{5}{4} \Rightarrow 4d = 5(x-2) \quad \dots (ii)$$

$$\Rightarrow 4d = 5x - 10 \quad \dots (ii)$$

From (i) and (ii)

$$3x + 6 = 5x - 10 \Rightarrow x = 8 \text{ km/h}$$

$$7. (a); \text{ Given } y = 2 \text{ km/hr}$$

$$\frac{10}{x-y} + \frac{10}{x+y} = \frac{55}{60} \Rightarrow 10 \left(\frac{x+y+x-y}{(x-y)(x+y)} \right) = \frac{11}{12}$$

$$\frac{20x}{x^2 - y^2} = \frac{11}{12} \Rightarrow \frac{20x}{x^2 - 4} = \frac{11}{12}$$

$$240x = 11x^2 - 44 \Rightarrow 11x^2 - 240x - 44 = 0$$

$$11x^2 - 242x + 2x - 44 = 0$$

$$11x(x-22) + 2(x-22) = 0$$

$$(11x+2)(x-22) = 0 \Rightarrow x = 22 \text{ km/h}$$

$$8. (a); \text{ Let speed in still water } = x, \text{ speed of current } = y$$

$$\text{Then, } \frac{55}{x+y} + \frac{40}{x-y} = 13 \quad \dots (i)$$

$$\frac{44}{x+y} + \frac{30}{x-y} = 10 \quad \dots (ii)$$

$$v = x + y, u = x - y$$

$$\frac{55}{v} + \frac{40}{u} = 13 \quad \dots (iii)$$

$$\frac{44}{v} + \frac{30}{u} = 10 \quad \dots (iv)$$

$$\text{From eq. (iii)} \times 3 - \text{eq. (iv)} \times 4$$

$$\frac{165}{v} + \frac{120}{u} = 39, \quad \frac{176}{v} + \frac{120}{u} = 40$$

$$\frac{11}{v} = -1 \Rightarrow v = 11, u = 5 \Rightarrow x + y = 11$$

$$x - y = 5$$

$$\text{Speed of man, } x = 8 \Rightarrow \text{Speed of current, } y = 3$$

$$9. (c); \text{ Speed of boat } = 10 \text{ km/h}$$

$$\text{Speed of stream } = 4 \text{ km/h}$$

$$\frac{d}{10+4} + \frac{d}{10-4} = 5, \quad \frac{d}{14} + \frac{d}{6} = 5, \quad \frac{3d+7d}{42} = 5,$$

$$d = \frac{5 \times 42}{10} = 21 \text{ km}$$

$$10. (a); \frac{24}{u} + \frac{36}{v} = 6 \quad \dots (i)$$

$$\frac{36}{u} + \frac{24}{v} = \frac{13}{2} \quad \dots (ii)$$

$$\text{eq. (i)} \times 3 - \text{eq. (ii)} \times 2$$

$$\frac{72}{u} + \frac{108}{v} = 18 \Rightarrow \frac{72}{u} + \frac{48}{v} = 13 \Rightarrow \frac{60}{v} = 5$$

$$v = 12, u = 8 \Rightarrow x + y = 12$$

$$x - y = 8 \Rightarrow x = 10, y = 2$$

$$\text{Velocity of current } = 2 \text{ km/hr.}$$

$$11. (c); \text{ Given, } x = 10 \text{ km/hr}$$

$$\frac{26}{x+y} = \frac{14}{x-y} \Rightarrow 13x - 13y = 7x + 7y$$

$$6x = 20y \Rightarrow 20y = 6 \times 10 \Rightarrow y = 3 \text{ km/hr}$$

$$12. (a); \text{ Given } y = 2 \text{ km/h}$$

$$\frac{6}{x-2} + \frac{6}{x+2} = \frac{33}{60} \Rightarrow 6 \left(\frac{2x}{x^2 - 4} \right) = \frac{33}{60}$$

$$11x^2 - 240x - 44 = 0$$

$$11x^2 - 242x + 2x - 44 = 0$$

$$(x-22)(11x+2) = 0 \Rightarrow x = 22 \text{ km/h}$$

$$13. (c); \text{ Given, } x = \frac{28}{3} \text{ km/h, speed of current } = y \text{ km/h}$$

$$\frac{d}{\frac{28}{3} - y} = \frac{3d}{\frac{28}{3} + y} \Rightarrow \frac{28}{3} + y = 28 - 3y$$

$$4y = \frac{56}{3}, y = \frac{14}{3}, y = 4\frac{2}{3} \text{ km/h}$$

$$14. (b); \text{ Given } x = 18 \text{ km/h}$$

$$\frac{d}{18-y} = \frac{3d}{18+y} \Rightarrow 18 + y = 3(18 - y)$$

$$18 + y = 54 - 3y \Rightarrow 4y = 36, y = 9 \text{ km/h}$$

$$15. (d); \frac{d}{x+y} = 8$$

$$d = 8x + 8y \quad \dots (i)$$

$$\frac{d}{x-y} = 12$$

$$d = 12x - 12y \quad \dots (ii)$$

$$\text{On solving (i) and (ii)}$$

$$8x + 8y = 12x - 12y \Rightarrow 4x = 20y, x = 5y$$

$$\text{Here } y = 5 \Rightarrow x = 5 \times 5 = 25 \text{ km/h}$$

16. (a); Speed of boat = x
Speed of river current = y

$$\frac{15}{x+y} = 3 \frac{45}{60} = \frac{15}{4}$$

$$x+y=4 \quad \dots (i)$$

$$\frac{5}{x-y} = 2 \frac{1}{2} = \frac{5}{2}$$

$$x-y=2 \quad \dots (ii)$$

On solving (i) and (ii)

$$x=3 \text{ and } y=1 \text{ km/h}$$

17. (c); $\frac{6}{x-y} + \frac{6}{x+y} = 2 \Rightarrow 6 \left(\frac{2x}{x^2 - y^2} \right) = 2$

Put $y = 4 \text{ km/hr}$

$$\frac{6x}{x^2 - 16} = 1 \Rightarrow x^2 - 16 = 6x \Rightarrow x^2 - 6x - 16 = 0$$

$$x^2 - 8x + 2x - 16 = 0 \Rightarrow x(x-8) + 2(x-8) = 0$$

$$(x+2)(x-8) = 0 \Rightarrow x = 8 \text{ km/hr}$$

18. (d); Here $x = 9 \text{ km/hr}$ and $y = 1.5 \text{ km/hr}$

$$\frac{105}{9+1.5} + \frac{105}{9-1.5} = T \Rightarrow T = \frac{105}{10.5} + \frac{105}{7.5}$$

$$= 10 + 14 = 24 \text{ hrs}$$

19. (c); $\frac{12}{x-y} + \frac{18}{x+y} = 3 \quad \dots (i)$

$$\frac{36}{x-y} + \frac{24}{x+y} = \frac{13}{2} \quad \dots (ii)$$

$$\text{Let } x-y=A, x+y=B$$

$$\frac{12}{A} + \frac{18}{B} = 3 \quad \dots (iii)$$

$$\frac{36}{A} + \frac{24}{B} = \frac{13}{2} \quad \dots (iv)$$

$$\text{Eq. (i)} \times 3 - \text{eq. (ii)}$$

$$\frac{36}{A} + \frac{54}{B} = 9 \Rightarrow \frac{36}{A} + \frac{24}{B} = \frac{13}{2}$$

$$\frac{30}{B} = 9 - \frac{13}{2} \Rightarrow \frac{30}{B} = \frac{5}{2} \Rightarrow B = 12$$

From eq. (iv)

$$\frac{36}{A} = \frac{9}{2} \Rightarrow A = 8 \Rightarrow x-y=8 \Rightarrow x+y=12$$

$$2x = 20 \Rightarrow x = 10 \text{ km/hr} \Rightarrow y = 2 \text{ km/hr}$$

20. (a); $\frac{48}{u} + \frac{48}{v} = 14 \quad \dots (i)$

$$\frac{3}{4} = \frac{u}{v} \Rightarrow u = \frac{3v}{4}$$

Put it in eq. (i)

$$\frac{4 \times 48}{3v} + \frac{48}{v} = 14 \Rightarrow \frac{7 \times 48}{3v} = 14$$

$$v = 8 \Rightarrow u = \frac{3}{4} \times 8 = 6 \Rightarrow x-y=6, x+y=8$$

$$x = 7 \Rightarrow y = 7 - 6 = 1 \text{ km/hr}$$

rate of stream = 1 km/hr

Previous Year (Memory Based)

1. (c); Down stream speed S_D = Speed of Boat + Speed of stream

$$S_B + S_S = 11$$

$$\frac{S_B - S_S = 5}{2S_B = 16}$$

$$\text{Speed of Boat} = 8 \text{ km/hr}$$

2. (b); Downstream speed of Boat = $\frac{16}{2} = 8 \text{ km/h}$

$$\text{Upstream speed} = \frac{16}{4} = 4 \text{ km/hr}$$

$$S_B + S_S = 8 \Rightarrow S_B - S_S = 4$$

from these two equations

$$S_B = 6 \Rightarrow S_S = 2$$

so speed of Boat in still water = 6 km/hr

3. (c); Boat man goes down stream in 1 hour = $1 \times 6 = 6 \text{ km}$

Downstream speed = 6 km/hr

Upstream speed = 2 km/hr

Speed of Boat in still water = 4 km/hr

$$\text{Time taken by Boat to cover } 5 \text{ km} = \frac{5}{4} \text{ hr.}$$

$$= 1 \text{ hr } 15 \text{ min}$$

4. (b); This type of question can also be done by option.

take (b) option so speeds are 3 and 1 km/hr

downstream distance cover in 1 hour = 4 km.

upstream distance = 2 km

By General Method:

Let's Man's upstream's speed is x km/hr

Downstream's speed = 2x km/h

In one hour distance covered in downstream

$$= 2x + x = 3x$$

In one hour upstream distance covered = $2x - x$
 $= x$

$$\frac{\text{Downstream's speed}}{\text{Upstream's speed}} = \frac{3x}{x} = 3 : 1$$

5. (c); Let's man upstream speed be x km/hr
 and downstream = y km/h
 Then Distance covered upstream in 8 hr 48 min
 = Distance covered downstream in 4 hrs.

$$x \times 8\frac{4}{5} = y \times 4 \Rightarrow \frac{44x}{5} = 4y$$

$$y = \frac{11}{5}x \Rightarrow \frac{y}{x} = \frac{11}{5}$$

$$\text{Required Ratio} = \left(\frac{y+x}{2}\right) : \left(\frac{y-x}{2}\right) = \frac{8}{3} = 8 : 3$$

6. (c); Speed of Boat upstream = $\frac{7}{42} \times 60 = 10$ km/hr.

$$\text{Speed of Boat} = 10 + (\text{Speed of stream}) \\ = 13 \text{ km/hr}$$

7. (a); Man's upstream speed = $5 - 1 = 4$ km/hr
 Down stream speed = 6 km/hr
 Let the distance is D km/h

$$\frac{D}{4} + \frac{D}{6} = 1 \Rightarrow \frac{3D + 2D}{12} = 1$$

$$D = \frac{12}{5} = 2.4 \text{ km}$$

8. (b); Let the speed of stream = S km/hr

$$\frac{30}{15-S} + \frac{30}{15+S} = \frac{9}{2}$$

From here try to use the option.

So option (b) i.e. $S = 5$ km/hr fit correctly.

9. (a); Let the speed of the stream be x mph,
 Downstream speed = $(10 + x)$ mph
 Upstream speed = $(10 - x)$ mph

$$\therefore \frac{36}{(10-x)} - \frac{36}{(10+x)} = \frac{90}{60}$$

$$x^2 + 48x - 100 = 0$$

$$(x + 50)(x - 2) = 0$$

$$x = 2 \text{ mph.}$$

10. (d); Let the speed in still water be x mph and the
 speed of the current be y mph. Then
 Upstream speed = $(x - y)$
 Downstream speed = $(x + y)$

$$\therefore \frac{12}{(x-y)} - \frac{12}{(x+y)} = 6$$

$$6(x^2 - y^2) = 24y$$

$$x^2 - y^2 = 4y$$

$$x^2 = (4y + y^2) \quad \dots(i)$$

$$\text{And } \frac{12}{(2x-y)} - \frac{12}{(2x+y)} = 1$$

$$x^2 = \frac{24y + y^2}{4} \quad \dots(ii)$$

From (i) and (ii) we have:

$$4y + y^2 = \frac{24y + y^2}{4}$$

$$3y^2 = 8y$$

$$y = \frac{8}{3} \quad \therefore \text{speed of current} = 2\frac{2}{3} \text{ mph}$$

11. (a); Let the speed of motorboat and stream is x km/h
 and y km/h respectively.

According to question

$$2\left(\frac{28}{x+y}\right) = \frac{28}{x-y}$$

$$x = 3y$$

when the speed of the stream is doubled

$$\frac{28}{x+2y} + \frac{28}{x-2y} = \frac{672}{60}$$

$$\frac{28}{5y} + \frac{28}{y} = \frac{672}{60} \quad [x = 3y]$$

$$\frac{1}{5y} + \frac{1}{y} = \frac{24}{60}$$

$$y = 3 \text{ km/h, } x = 9 \text{ km/h}$$

12. (b); Let the speed of B = x km/hr

Then the speed of A = $(x + 5)$ km/hr

$\therefore$ Distance travelled by A to meet his friend
 $= 42 \text{ km}$

Distance travelled by B = $35\frac{5}{7}\%$ less of A

$$= \left[\frac{250}{7 \times 100} = \frac{5}{14} \right]$$

Distance travelled by A = $14x$

by B = $9x$ ($14 - 5 = 9$)

$$14x = 42, \quad x = 3$$

$$9 \times 3 = 27 \text{ km} \Rightarrow \text{travelled by B.}$$

Time is equal

$$\frac{42}{x+5} = \frac{27}{x} \Rightarrow x = 9 \text{ km/hr}$$

Speed by A = 9 + 5 = 14 km/hr

13. (b); Let the distance between Varansi and Patna is d km, and the speed of the boat in still water is x km/h, and the speed of stream is y km/h.

∴ Diya will move at a speed of stream

$$24 = \frac{d}{y} \Rightarrow d = 24y \text{ km}$$

$$\therefore \frac{24y}{x+y} + \frac{24y}{x-y} = 10$$

When speed of boat increases by 40%

$$\frac{24y}{\frac{7}{5}x+y} + \frac{24y}{\frac{7}{5}x-y} = 7$$

After solving this we get
x = 5y

$$t = \frac{24y}{6y} \left[\begin{array}{l} \text{time taken to sail from} \\ \text{Varansi to Patna} \end{array} \right] \\ = 4 \text{ hours}$$

14. (a); Let the distance between two points A and B is d km. And the speed of motoboat in still water = x km/hr
speed of stream is y km/h

$$\text{From question } \frac{d}{x+y} + \frac{d}{x-y} = 5$$

Let original time = 5

20% of 5 = 1 (new time)

$$d(x+y+x-y) = 5(x^2-y^2)$$

$$d = \frac{5(x^2-y^2)}{2x} \quad \dots(i)$$

When the speed of Boat is doubled

$$\frac{d}{2x+y} + \frac{d}{2x-y} = 1$$

$$d[2x+y+2x-y] = 4x^2-y^2$$

$$d = \frac{4x^2-y^2}{4x} \quad \dots(ii)$$

d is same in both the cases.

$$\frac{5(x^2-y^2)}{2x} = \frac{4x^2-y^2}{4x}$$

$$6x^2 = 9y^2$$

$$\frac{x}{y} = \sqrt{\frac{3}{2}}$$

15. (c); River : Boat

$$\text{Speed } v : 4v$$

Upstream speed = 3v

Distance = t × speed

$$= 3v \times 2 = 6v$$

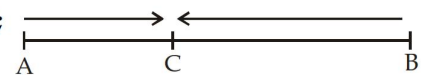
Downstream speed = 5v

$$\text{Speed} = \frac{D}{T} = \frac{6v}{\left(1+\frac{1}{2}\right)} = 4v$$

Initial Downstream speed = 5v

Final Downstream speed = 4v

$$\% \text{ decrease} = \frac{5v-4v}{5v} \times 100 = 20\%$$

16. (b); 

speed of boat A = b_1

speed of boat B = b_2

let they meet at point C

$$\text{Total time} = \frac{\text{total distance}}{\text{Relative speed}} = \frac{2W}{b_1+b_2}$$

$$\text{Distance travelled by A} = \frac{2W}{b_1+b_2} \times b_1$$

$$\text{Distance travelled by B} = \frac{2W}{b_1+b_2} \times b_2$$

17. (a); Let the speed of Boat in still water = x km/h
and that of current = y km/h

$$\frac{60}{x+y} + \frac{60}{x-y} = 10 \quad \dots(i)$$

$$\frac{3}{x+y} = \frac{2}{x-y}$$

$$3(x-y) = 2(x+y)$$

$$\frac{x+y}{x-y} = \frac{3}{2}$$

put these value in (i)

$$\frac{60}{3R} + \frac{60}{2R} = 10$$

$$R = 5, \text{ then } x+y = 15$$

$$x-y = 10$$

$$\text{speed of Boat} = \frac{1}{2}(15+10) = 12.5 \text{ km/h}$$

$$\text{speed of current} = 2.5 \text{ km/h}$$

18. (d); Upstream speed = $\frac{3}{4} \times \frac{1}{15} \times 60 = 3 \text{ km/hr}$

Downstream speed = $\frac{3}{4 \times 10} \times 60 = 4.5 \text{ km/h}$

speed of boat = $\frac{3 + 4.5}{2} = 3.75 \text{ km / h}$

speed of current = $\frac{4.5 - 3}{2} = .75 \text{ km / h}$

Ratio of his speed to current = $\frac{3.75}{.75} = 5 : 1$

19. (c); speed of Boat : speed of current = 36 : 5
so, downstream speed = 41

Distance = $41 \times \frac{1}{6}$

upstream speed = 31

Time = $\frac{41}{6 \times 31} = \frac{41}{186} \text{ hr.} = 13.22 \text{ min.}$

20. (a); Downstream speed = 20 km/h

Upstream speed = $\frac{20}{2} = 10 \text{ km / h}$

Speed of Boat = $\frac{20 + 10}{2} = 15 \text{ km / h}$